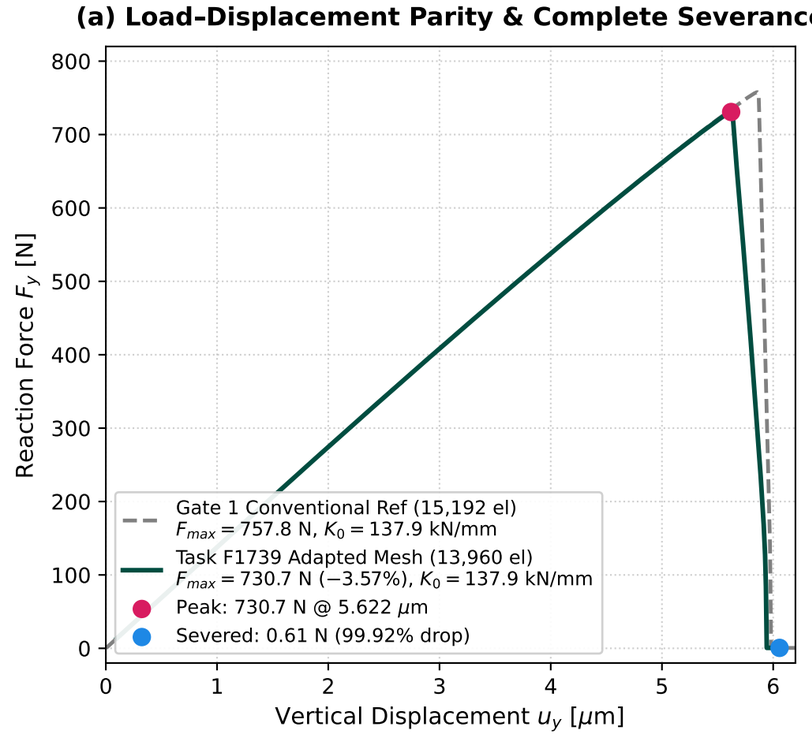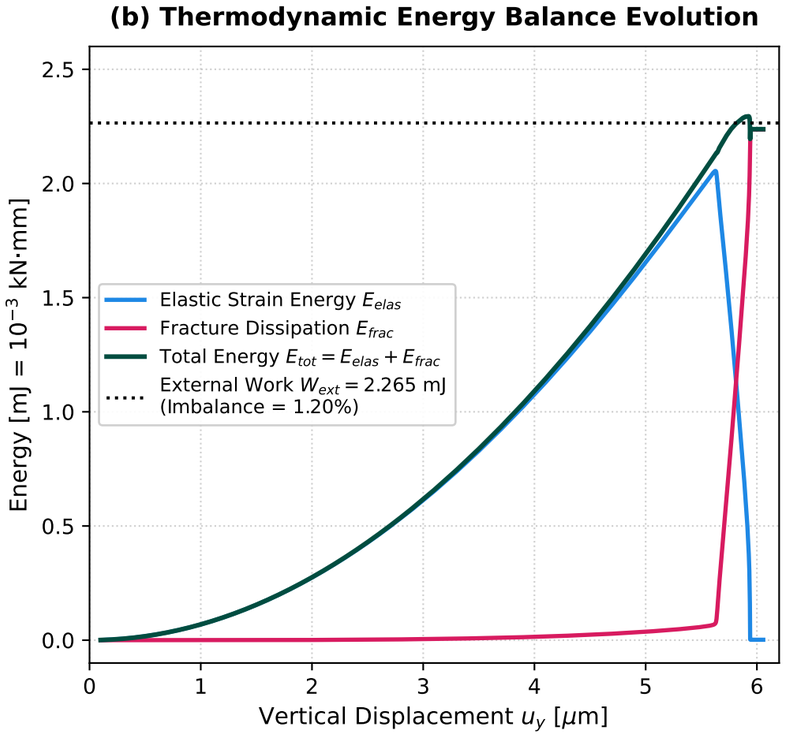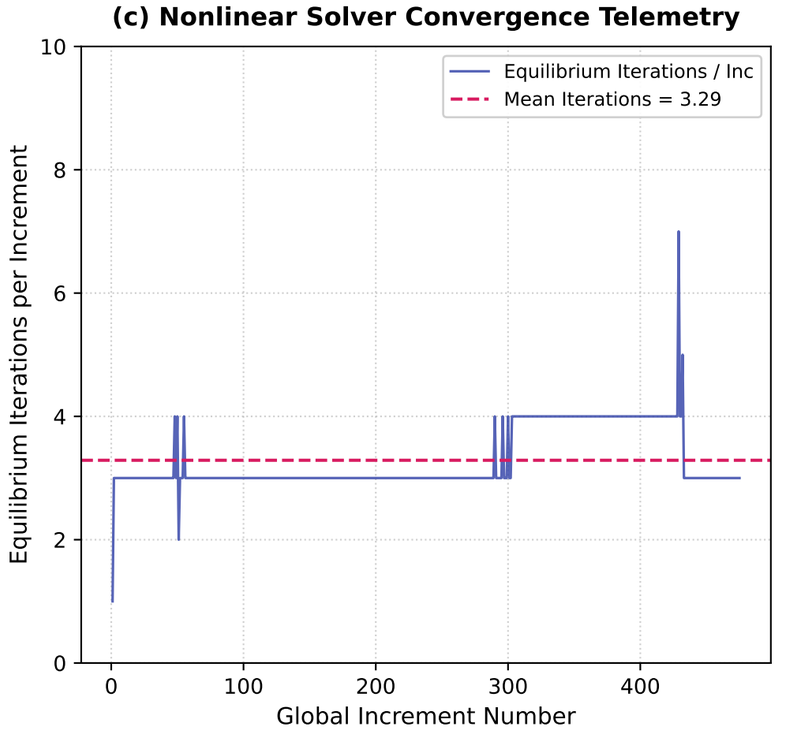Task F1739: Physics-Driven Mode-I Coupled Fracture Simulation on Ligament-Qualified Adapted Mesh